1.Creation of Amazon EC2 Instances (Linux and Windows) and Establishing Secure Connections Using SSH from Local Machine, PuTTY (.pem & .ppk), AWS CloudShell, and Remote Desktop Protocol (RDP)

□ PREREQUISITES

Make sure you have:

- ✅ EC2 **Ubuntu** instance running
- ✅ **.ppk private key file**
- ✅ **PuTTY installed (Windows)-- https://putty.org/index.html (INSTALL FIRST ONE)**

◈ STEP 1: Check Security Group

Go to **EC2 → Security Groups → Inbound rules**

Ensure this rule exists:

Type: SSH

Port: 22

Source: My IP

❌ If SSH is blocked → connection will fail

◈ STEP 2: Open PuTTY

1. Launch **PuTTY**
2. In **Session** screen:

Host Name (or IP address): EC2—CONNECT—COPY (DNS NAME)

ubuntu@<PUBLIC-IP>

Example:

ubuntu@3.110.xxx.xxx

◈ STEP 3: Attach the .ppk Key

In PuTTY left menu, go to:

Connection → SSH → Auth → Credentials

1. Click **Browse**
2. Select your **.ppk file**
3. Click **Open**

◈ STEP 4: Connect to EC2

Click **Open**

- First time → click **Yes** (security alert)
- PuTTY will log in automatically as:

ubuntu

🎉 **You are now connected to your Ubuntu EC2 instance**

▯ VERIFY CONNECTION

Run:

```
whoami
```

Output should be:

```
Ubuntu
```

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

1_ppk

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

32be01f26a3de55 2026-01-22 ec2-user Verified provider

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

key_ppk

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

- (●) **RSA** — RSA encrypted private and public key pair
- ( ) **ED25519** — ED25519 encrypted private and public key pair

Private key file format

- ( ) **.pem** — For use with OpenSSH
- (●) **.ppk** — For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. **You will need it later to connect to your instance.** Learn more

Cancel | Create key pair

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0b4ed13277d59d862)

▸ Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

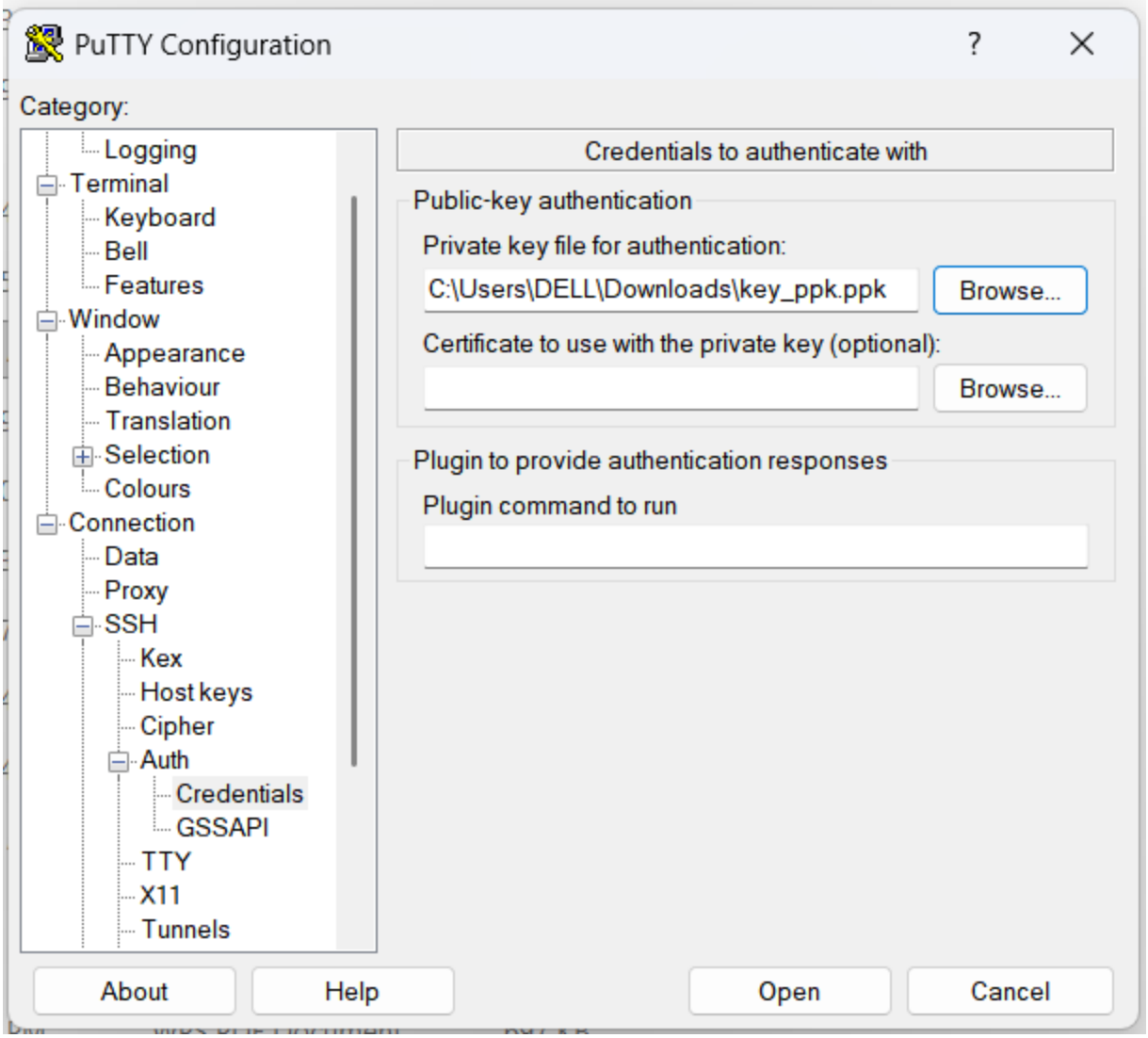
1 2

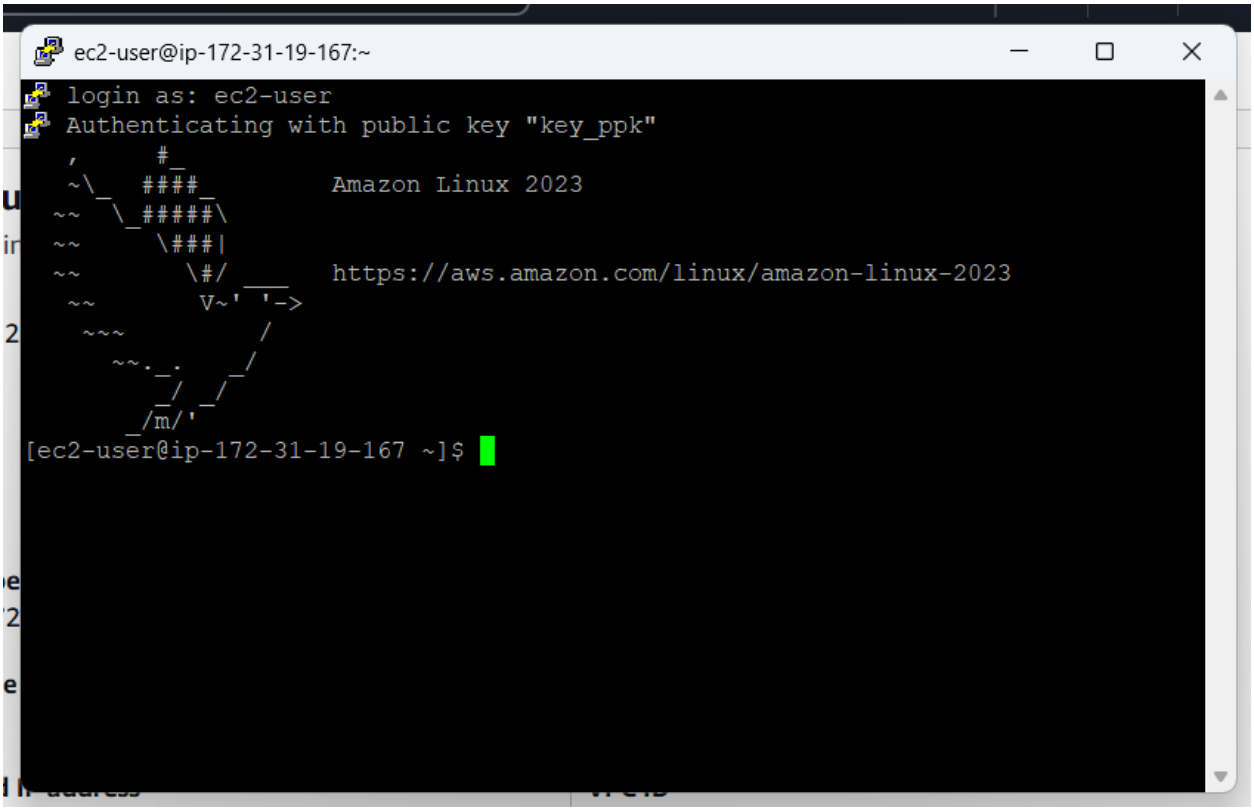
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts

Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database

Create EBS snapshot p
Create a policy that automate retention, and deletion of EBS
Create EBS snapshot poli


PuTTY Configuration
Category:
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
C:\Users\DELL\Downloads\key_ppk.ppk
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel



Connect EC2 Ubuntu Instance Using PuTTY (Windows) ---USING PEM KEYPAIR

PREREQUISITES

You must have:

- ✅ **EC2 Ubuntu instance running**
- ✅ **Public IPv4 address** (or Elastic IP)
- ✅ **Key pair (.pem file)** used while launching EC2

- ✅ **PuTTY & PuTTYgen installed**

👉 Download PuTTY:
https://www.putty.org

🔹 STEP 1: Convert .pem to .ppk using PuTTYgen

PuTTY **cannot use .pem directly**.

Steps:

1. Open **PuTTYgen**
2. Click **Load**
3. Select your .pem file
   (change file type to *All Files*)
4. Click **Open**
5. Click **Save private key**
6. Save as keyname.ppk

✓ Conversion done

🔹 STEP 2: Get EC2 Public IP

Go to:
AWS Console → EC2 → Instances---CONNECT –COPY DNS NAME

🔹 STEP 3: Configure PuTTY

1⃣ Open PuTTY

Session:

- **Host Name (or IP address): dns name**

2⃣ Load Private Key

Go to:

Connection → SSH → Auth → Credentials

- Click **Browse**
- Select your .ppk file

🔷 STEP 4: Connect

Click **Open**

- First time → click **Yes** (security alert)
- Login happens automatically as:

ubuntu

🎉 **You are now connected to Ubuntu EC2**

▫ VERIFY CONNECTION

Run:

```
whoami
```

Output:

```
ubuntu
```

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

1_pem

Add additional tags

Application and OS Images (Amazon Machine Image) Info

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Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-0532be01f26a3de55

Virtual server type (instance type)
t3.micro

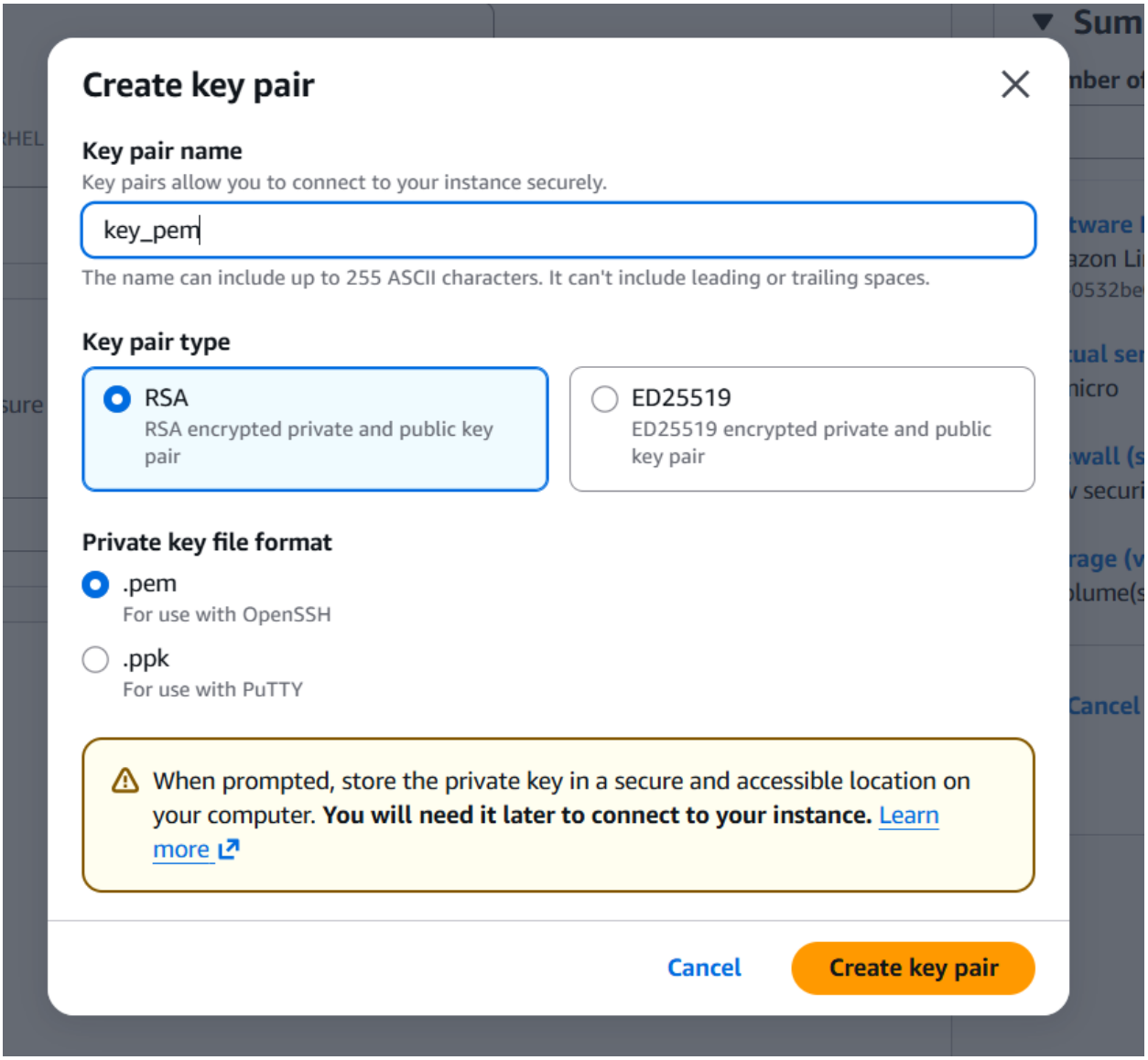
Firewall (security group)
New security group

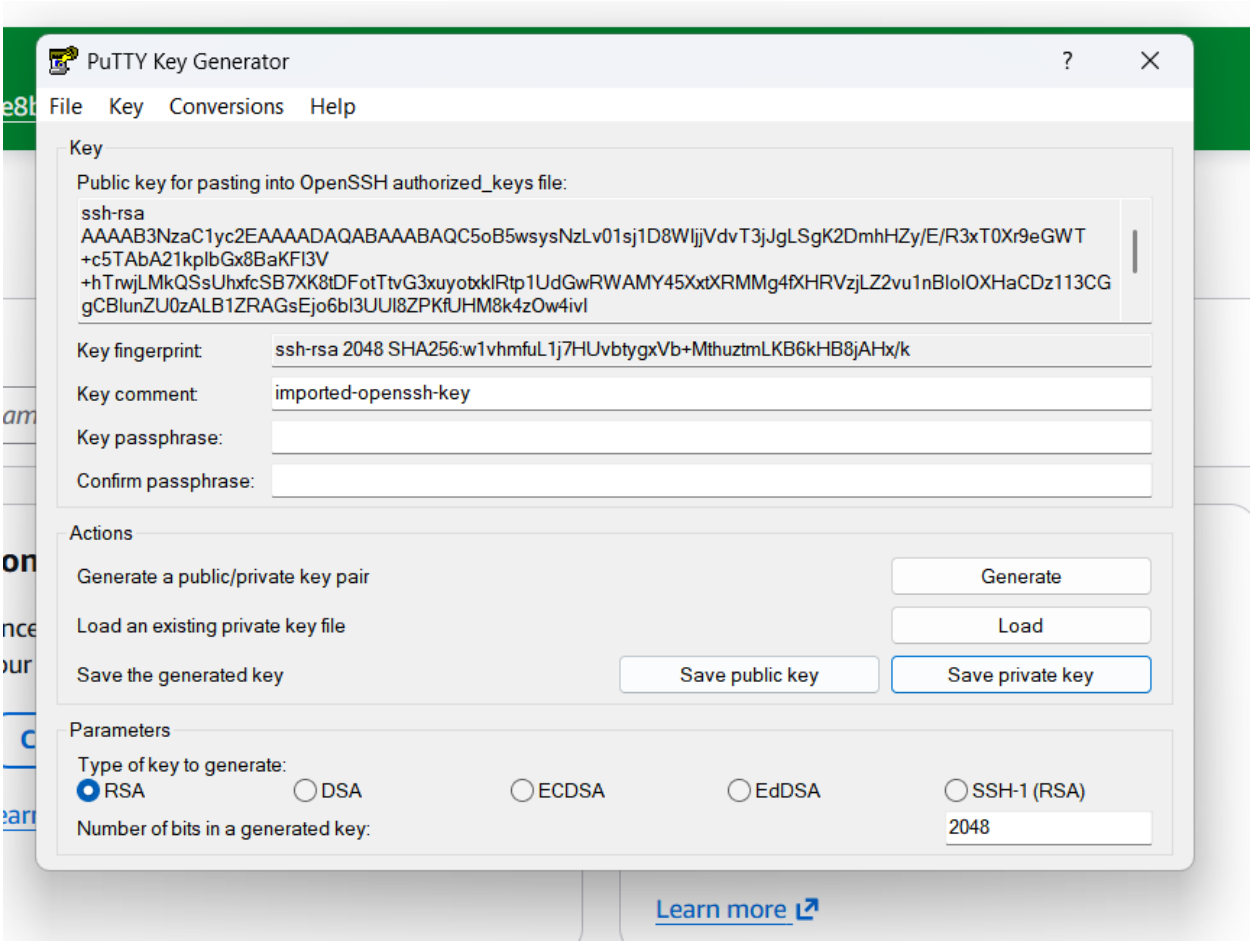
Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch i

Previ


Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
key_pem
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair


PuTTY Key Generator
File
Key
Conversions
Help
Key
Public key for pasting into OpenSSH authorized_keys file:
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAABAQC5oB5wsysNzLv01sj1D8WIjjVdvT3jJgLSgK2DmhHZy/E/R3xT0Xr9eGWT
+c5TAbA21kplbGx8BaKFI3V
+hTrwjLMkQSsUhxfcSB7XK8tDFotTtvG3xuyotxklRtp1UdGwRWAMY45XxtXRMMg4fXHRVzjLZ2vu1nBIoIOXHaCDz113CG
gCBlunZU0zALB1ZRAGsEjo6bI3UUI8ZPKfUHM8k4zOw4ivl
Key fingerprint:
ssh-rsa 2048 SHA256:w1vhmfuL1j7HUvbtygxVb+MthuztmLKB6kHB8jAHx/k
Key comment:
imported-openssh-key
Key passphrase:
Confirm passphrase:
Actions
Generate a public/private key pair
Generate
Load an existing private key file
Load
Save the generated key
Save public key
Save private key
Parameters
Type of key to generate:
RSA
DSA
ECDSA
EdDSA
SSH-1 (RSA)
Number of bits in a generated key:
2048
Learn more

Connect Info

Connect to an instance using the browser-based client.

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID

i-0df2ad703284e8bbb (1_pem)

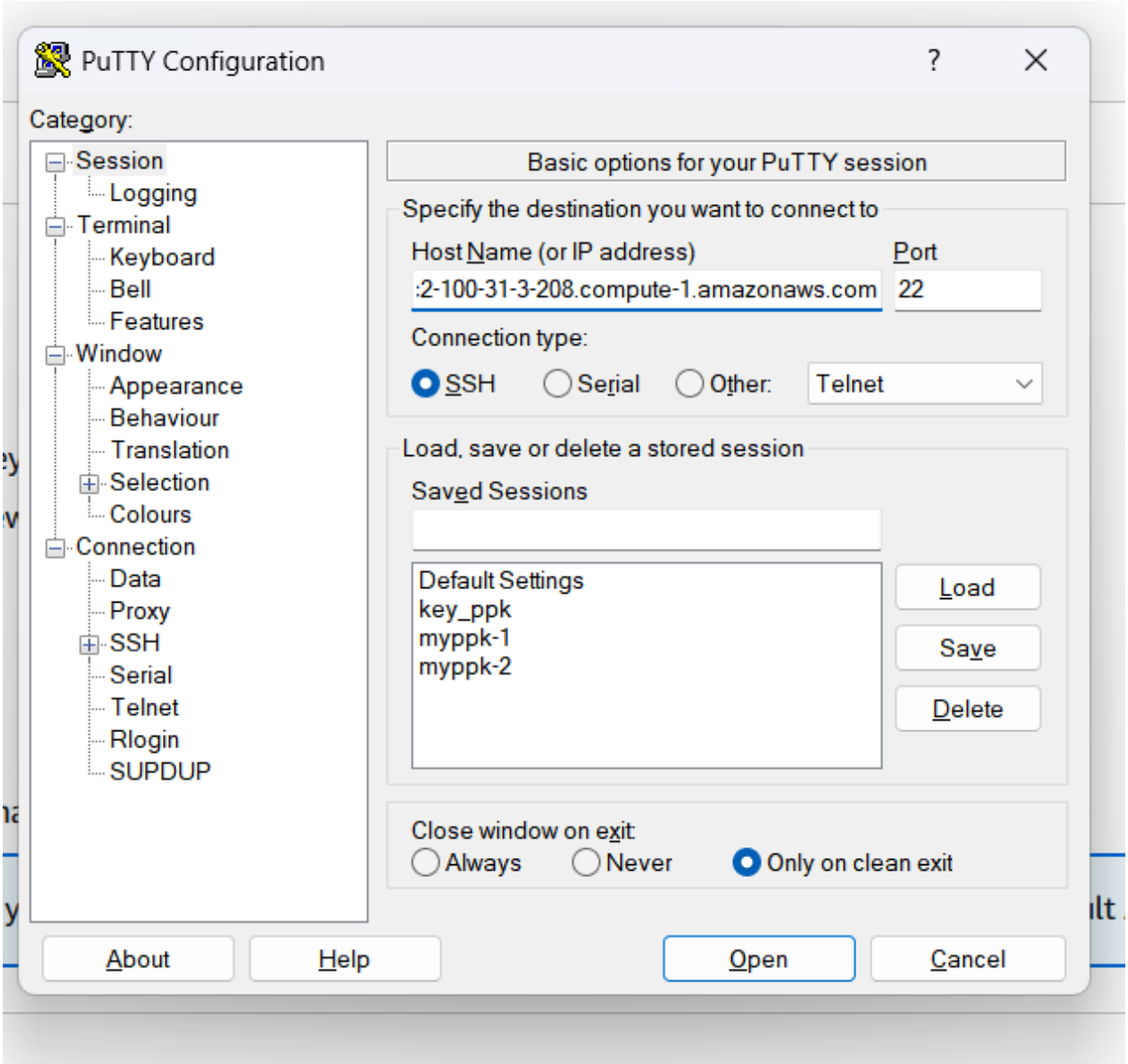
1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is key_pem.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
   "key_pem.pem"
4. Connect to your instance using its Public DNS:
   ec2-100-31-3-208.compute-1.amazonaws.com

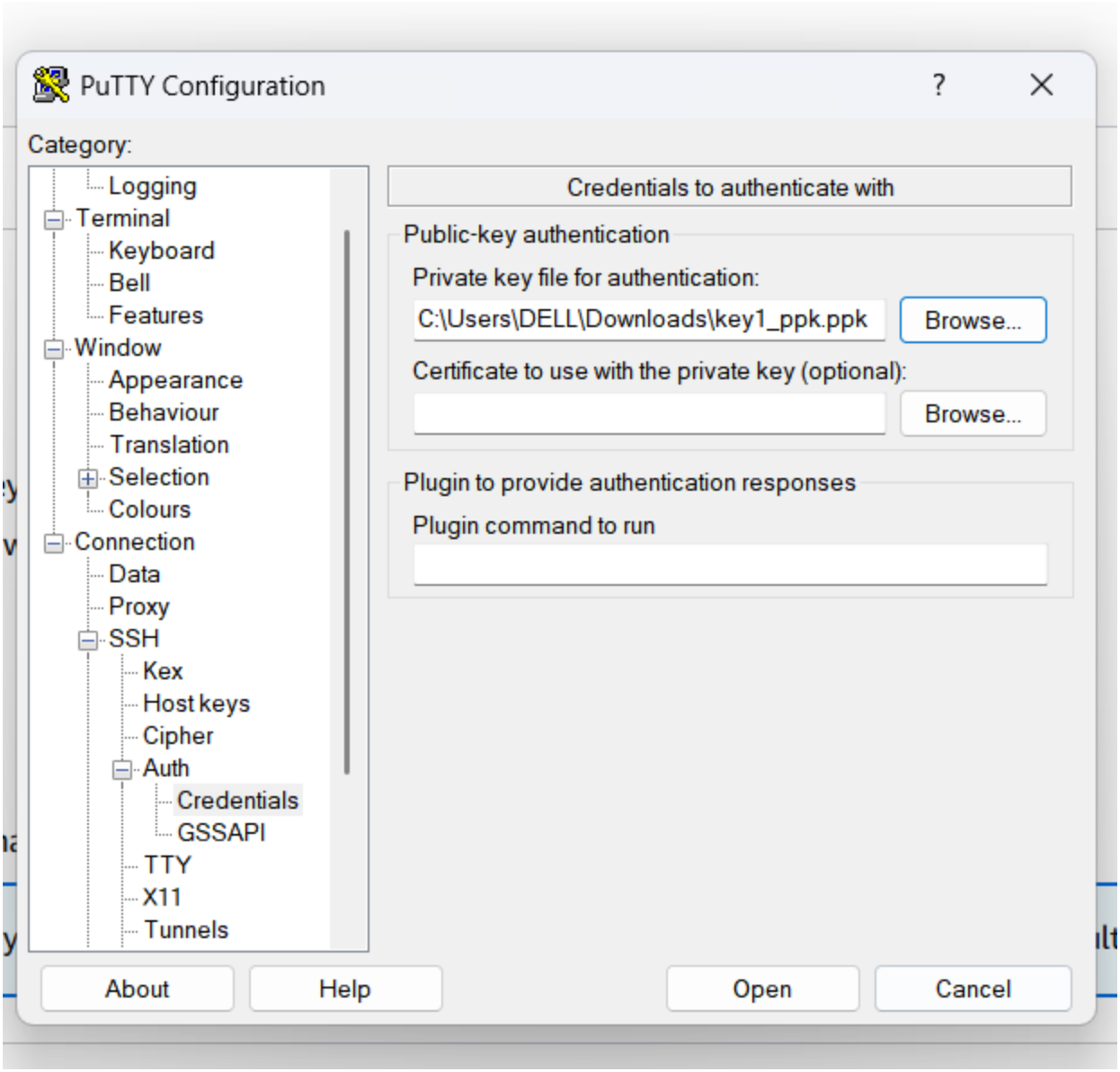
Public DNS copied

Example:

ssh -i "key_pem.pem" ec2-user@ec2-100-31-3-208.compute-1.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AM


PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Serial
Telnet
Rlogin
SUPDUP
Basic options for your PuTTY session
Specify the destination you want to connect to
Host Name (or IP address)
Port
:2-100-31-3-208.compute-1.amazonaws.com
22
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Default Settings
key_ppk
myppk-1
myppk-2
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel


PuTTY Configuration
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Features
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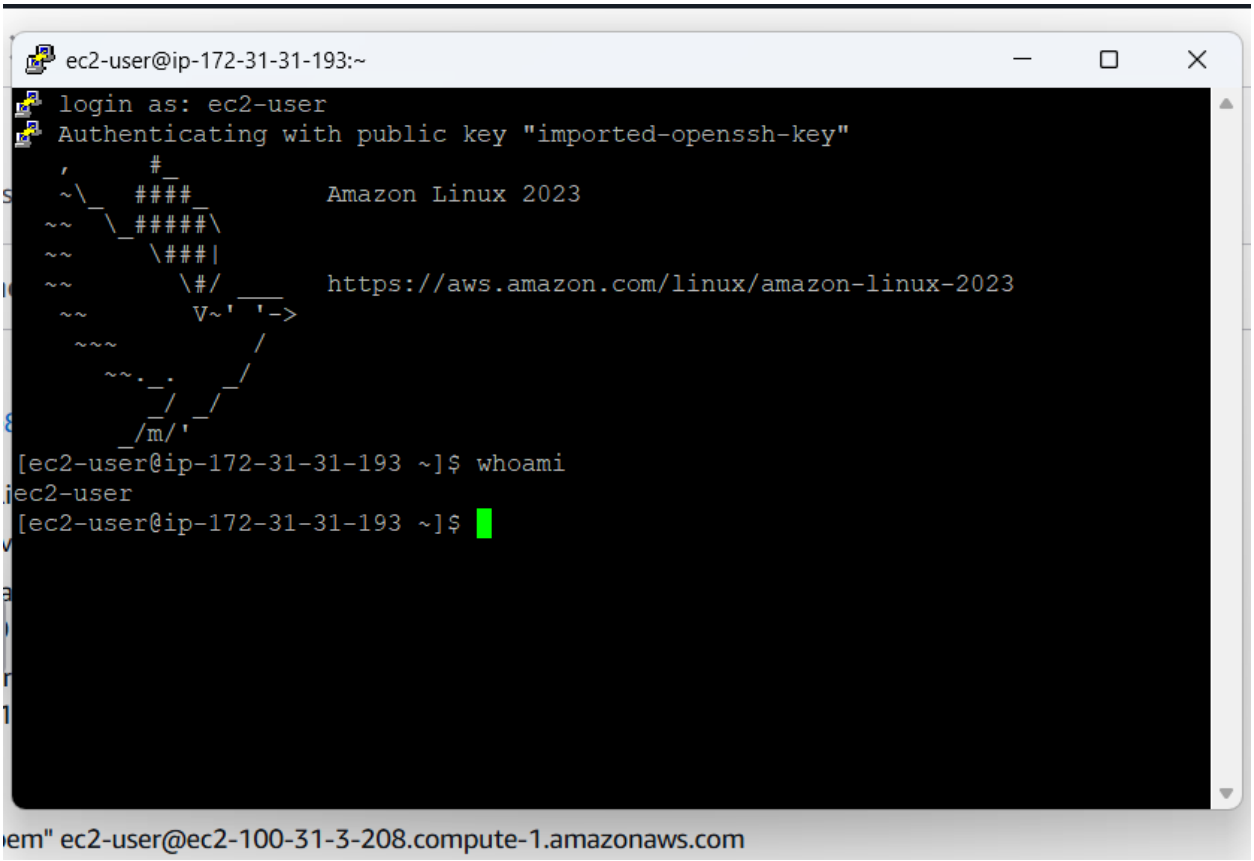
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
C:\Users\DELL\Downloads\key1_ppk.ppk
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel



Connecting to EC2 Using AWS CloudShell

1. **Create the EC2 instance** using a key pair (.pem).
2. **Log in to AWS Console** and click on the **CloudShell** icon (terminal icon on the top-right).
3. Once CloudShell opens, go to **AWS Services → EC2**.
4. Select your **EC2 instance** and click **Connect**.

5. Choose the **SSH client** tab.

6. **Copy the ssh -i command** shown there.

7. **Paste the command in CloudShell** and try to run it.
   ❌ It will fail with an error like:
   "key pair is not available / no such file or directory"
   (because the key file is not yet in CloudShell).

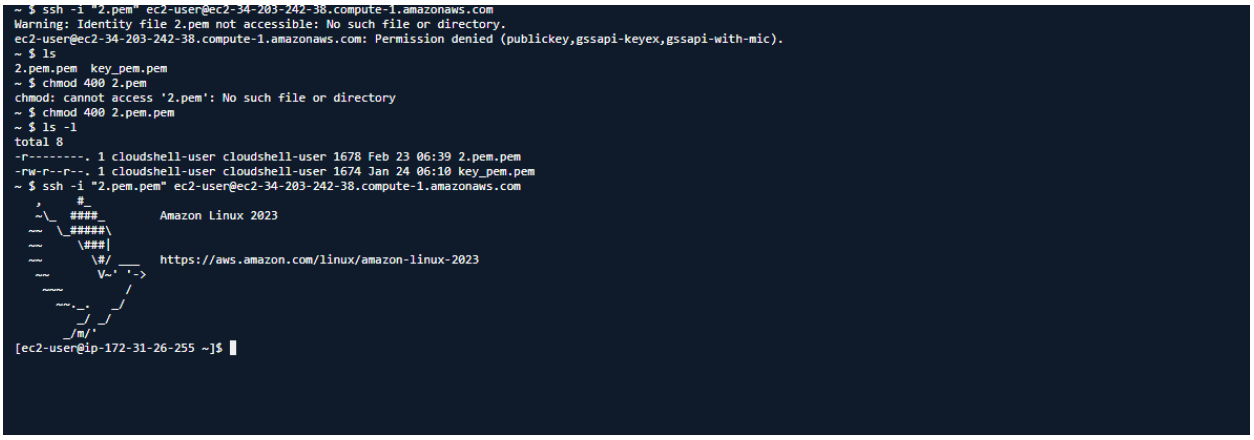
8. Now, in **CloudShell**, click on the **"+" (Actions)** icon on the right side.

9. Select **Upload file** and upload the **key pair (.pem)** from your local machine.

10. After upload, **copy the ssh -i command again** from **EC2 → Connect → SSH client**.

11. **Paste and execute the command** in CloudShell.

12. ✅ You are now **successfully connected to the EC2 instance**.



Step-by-Step: Create Windows EC2 Instance and Connect via RDP

Step 1: Launch a Windows EC2 Instance

1. Log in to your **AWS Management Console**.

2. Navigate to **EC2 → Instances → Launch Instances**.

3. Select an **Amazon Machine Image (AMI)**:

   - Choose a **Windows Server** AMI (e.g., Windows Server 2019 or 2022).

4. Choose an **Instance Type** (e.g., t2.micro for free tier or as per your requirement).
5. Configure **Instance Details** as needed.
6. Add **Storage** if required (default is usually sufficient).
7. Configure **Security Group**:
    - Ensure **RDP port 3389** is allowed.
    - Source can be **My IP** (for security) or a wider range if needed.
8. Select or **create a new Key Pair**:
    - Download the .pem file (keep it safe; it's needed to decrypt the password).
9. Click **Launch Instance**.

Step 2: Get the Public DNS / IP

1. After the instance is running, go to **EC2 → Instances**.
2. Select your instance and note the **Public DNS (IPv4)** or **Public IP**.

Step 3: Retrieve the Windows Administrator Password

1. Select your instance → Click **Connect → RDP Client**.
2. Click **Get Password**.
3. Upload your **Key Pair (.pem) file**.
4. Click **Decrypt Password**.
5. Copy the **Administrator password** shown.

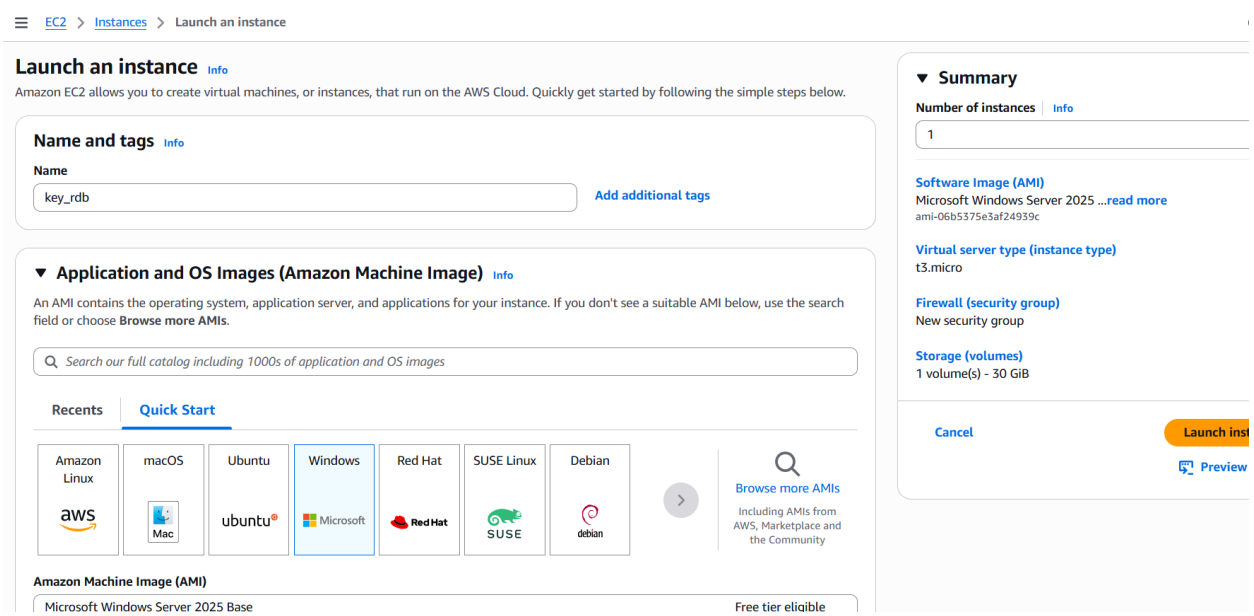
Step 4: Connect via RDP

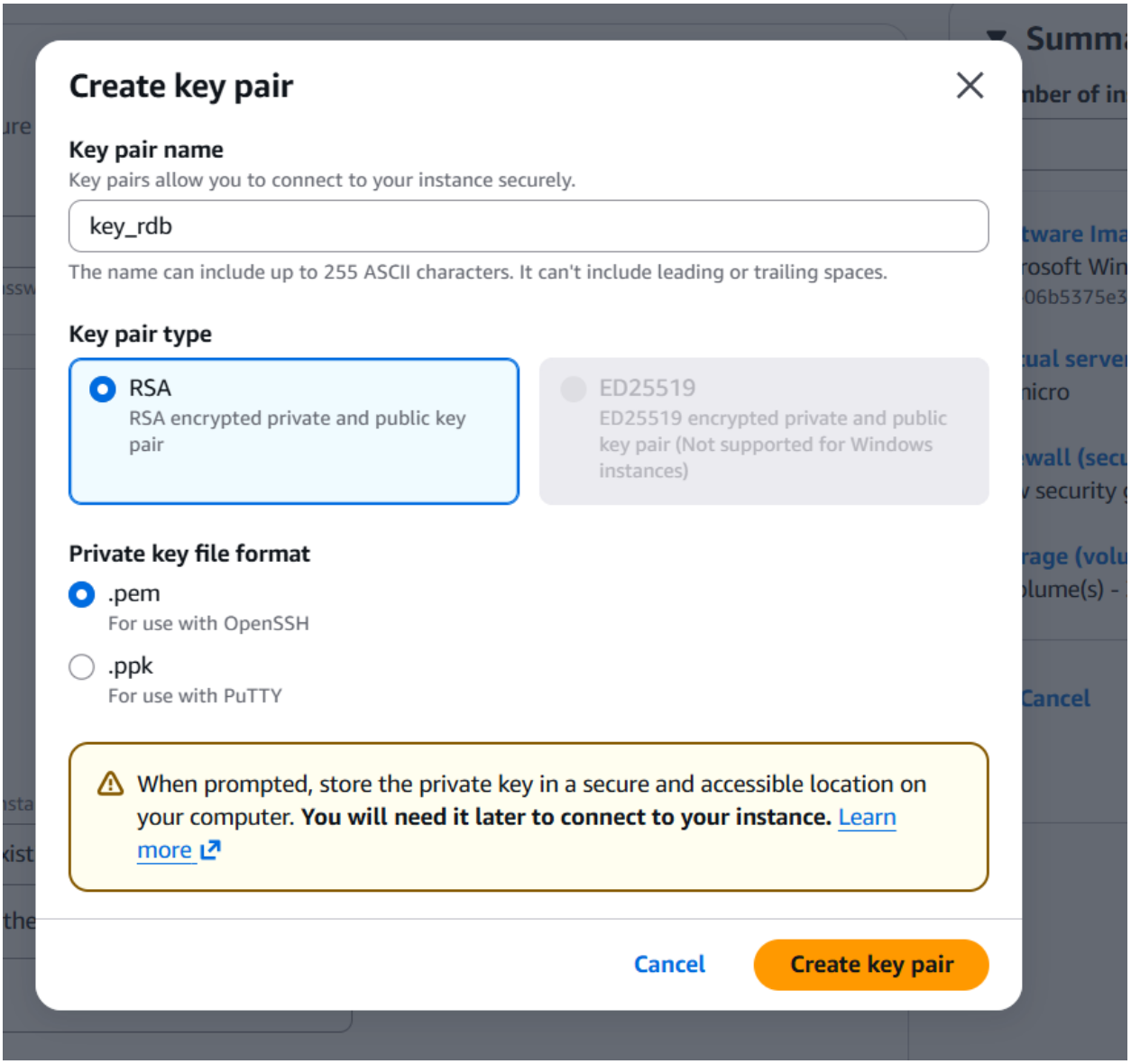
1. On your **local Windows machine**, open **Remote Desktop Connection** (mstsc).
2. In the **Computer** field, enter the **Public DNS** or **IP address** of the EC2 instance.
3. Click **Connect**.

4. In the login prompt:
    - **Username**: Administrator
    - **Password**: Paste the decrypted password from Step 3.
5. Click **OK**.

Step 5: You're connected

- Your Windows EC2 instance desktop should appear.
- You can now use it as a normal Windows machine.




Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
key_rdb
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair (Not supported for Windows instances)
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair

Firewall (security groups) | Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (●) Create security group
- ( ) Select existing security group

Security group name - *required*

launch-wizard-6

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* | Info

launch-wizard-6 created 2026-01-24T06:11:59.148Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 152.57.172.142/32) — Remove

Type	Protocol	Port range
rdp	TCP	3389

Source type	Name	Description - *optional*
My IP	Add CIDR, prefix list or security group 152.57.172.142/32 ✕	e.g. SSH for admin desktop

Add security group rule

Connect Info

Connect to an instance using the browser-based client.

Session Manager | RDP client | EC2 serial console

Record RDP connections
You can now record RDP connections using AWS Systems Manager just-in-time node access. Learn more

Instance ID
i-0a93cf4b5c0a6abdb (key_rdb)

Connection Type

- (●) Connect using RDP client
  Download a file to use with your RDP client and retrieve your password.
- ( ) Connect using Fleet Manager
  To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be in
  instance. For more information, see Working with SSM Agent

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

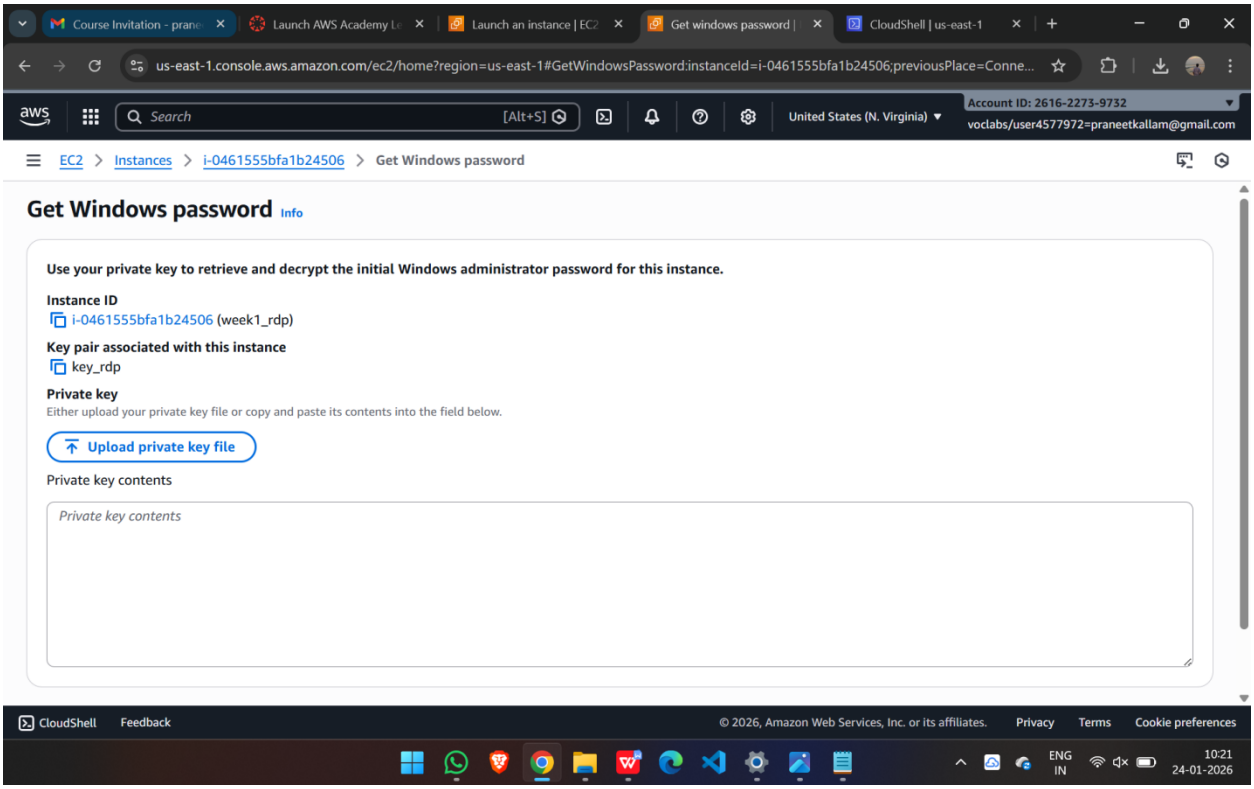
Download remote desktop file

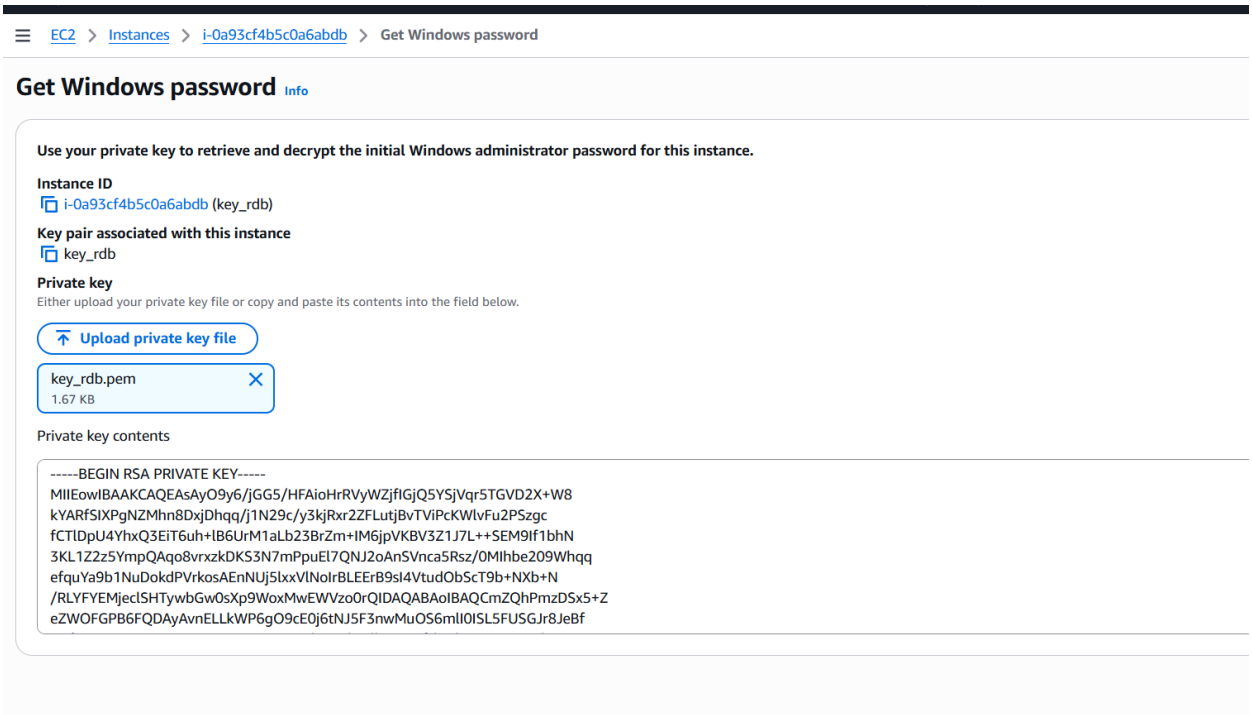
When prompted, connect to your instance using the following username and password:

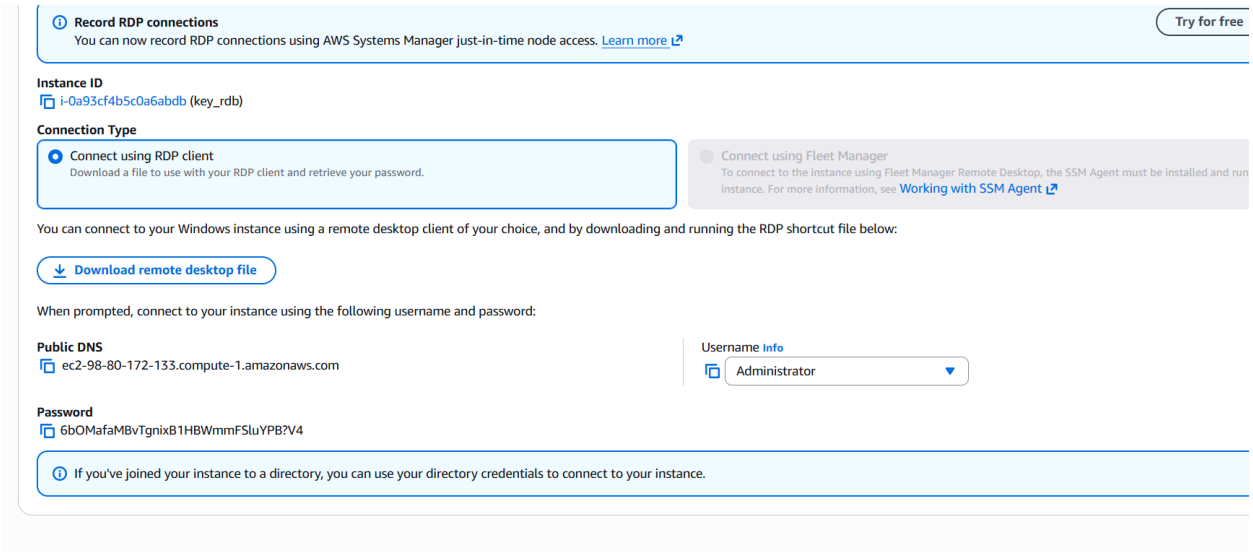
Public DNS
ec2-98-80-172-133.compute-1.amazonaws.com

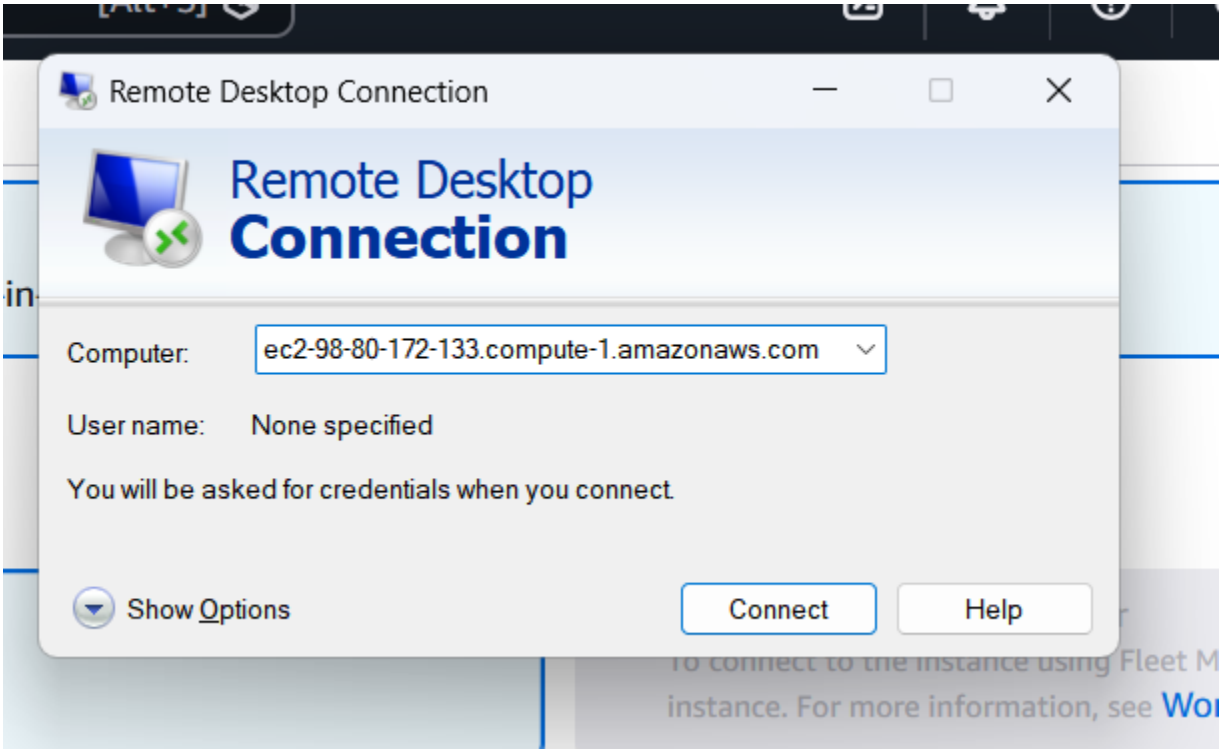
Username Info
Administrator

Password Get password


Course Invitation - prane
Launch AWS Academy Le
Launch an instance | EC2
Get windows password |
CloudShell | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#GetWindowsPassword:instanceId=i-0461555bfa1b24506;previousPlace=Conne...
aws
Search
[Alt+S]
United States (N. Virginia)
Account ID: 2616-2273-9732
voclabs/user4577972=praneetkallam@gmail.com
EC2 > Instances > i-0461555bfa1b24506 > Get Windows password
Get Windows password Info
Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.
Instance ID
i-0461555bfa1b24506 (week1_rdp)
Key pair associated with this instance
key_rdp
Private key
Either upload your private key file or copy and paste its contents into the field below.
Upload private key file
Private key contents
Private key contents
CloudShell
Feedback
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
ENG
IN
10:21
24-01-2026


EC2 > Instances > i-0a93cf4b5c0a6abdb > Get Windows password
Get Windows password Info
Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.
Instance ID
i-0a93cf4b5c0a6abdb (key_rdb)
Key pair associated with this instance
key_rdb
Private key
Either upload your private key file or copy and paste its contents into the field below.
Upload private key file
key_rdb.pem
1.67 KB
Private key contents
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAsAyO9y6/jGG5/HFAioHrRVyWZjfIGjQ5YSjVqr5TGVD2X+W8
kYARfSIXPgNZMhn8DxjDhqq/j1N29c/y3kjRxr2ZFLutjBvTViPcKWlvFu2PSzgc
fCTlDpU4YhxQ3EiT6uh+lB6UrM1aLb23BrZm+IM6jpVKBV3Z1J7L++SEM9If1bhN
3KL1Z2z5YmpQAqo8vrxzkDKS3N7mPpuEl7QNJ2oAnSVnca5Rsz/0MIhbe209Whqq
efquYa9b1NuDokdPVrkosAEnNUj5lxxVlNoIrBLEErB9sI4VtudObScT9b+NXb+N
/RLYFYEMjeclSHTywbGw0sXp9WoxMwEWVzo0rQIDAQABAoIBAQCmZQhPmzDSx5+Z
eZWOFGPB6FQDAyAvnELLkWP6gO9cE0j6tNJ5F3nwMuOS6mlI0ISL5FUSGJr8JeBf


Record RDP connections
You can now record RDP connections using AWS Systems Manager just-in-time node access. Learn more
Try for free
Instance ID
i-0a93cf4b5c0a6abdb (key_rdb)
Connection Type
Connect using RDP client
Download a file to use with your RDP client and retrieve your password.
Connect using Fleet Manager
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and run
instance. For more information, see Working with SSM Agent
You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:
Download remote desktop file
When prompted, connect to your instance using the following username and password:
Public DNS
ec2-98-80-172-133.compute-1.amazonaws.com
Username Info
Administrator
Password
6bOMafaMBvTgnixB1HBWmmFSluYPB?V4
If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.


Remote Desktop Connection
Remote Desktop
Connection
Computer:
ec2-98-80-172-133.compute-1.amazonaws.com
User name:
None specified
You will be asked for credentials when you connect.
Show Options
Connect
Help

VIVA QUESTIONS:

1)What is AWS EC2 Instance?

A)An AWS EC2 (Elastic Compute Cloud) instance is a virtual server in the AWS cloud used to run applications.

It provides computing resources like CPU, memory (RAM), storage, and operating system on a pay-as-you-go basis.

2)Components of EC2

A) i)AMI (Amazon Machine Image):Template used to launch an instance

ii) Instance Type:Defines CPU, RAM, network performance

Examples: t2.micro – low workload

iii)Storage:EBS (Elastic Block Store) – persistent storage

Instance Store – temporary storage

iv) Key Pair :Used for secure login (SSH/RDP)

Public key stored in AWS

Private key stored by user

v) Security Group:Acts as a virtual firewall ,Controls inbound and outbound traffic

Example: Allow HTTP (80), SSH (22)

vi) Network (VPC):EC2 runs inside a Virtual Private Cloud

Includes:Subnet

IP address (public/private)

Internet Gateway

3) How EC2 Works (Step-by-Step)

A) User selects an AMI

Chooses instance type

Attaches storage (EBS)

Configures security group

Selects key pair

Instance is launched

User connects via SSH or RDP

Application runs on EC2

4) Uses of EC2

A) Hosting websites

Running backend servers

Database servers

Application development & testing

Machine learning workloads

Enterprise applications

Batch processing

5)Advantages of EC2

A)Scalable – Increase/decrease resources anytime

Cost-effective – Pay only for usage

Flexible – Multiple OS and instance types

Highly secure – IAM, security groups, encryption

High availability – Multiple availability zones

Easy backup – EBS snapshots